

1. In the figure, a crate of mass $m = 100 \text{ kg}$ is pushed at constant speed up a frictionless ramp ($\theta = 30.0^\circ$) by a horizontal force $\vec{F}$.

(a) What is the magnitude of $\vec{F}$?

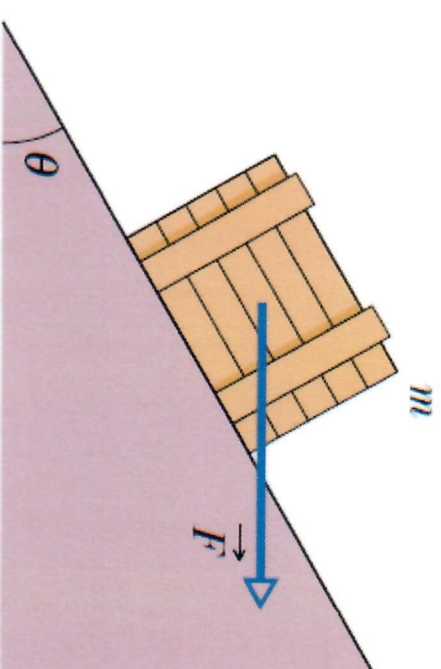
(b) What is the magnitude of the force on the crate from the ramp?

Die figuur dui 'n krat aan van massa $m = 100 \text{ kg}$ wat teen 'n konstante spoed by 'n wrywinglose skuinsvlak opbeweeg as gevolg van 'n horisontale krag $\vec{F}$.

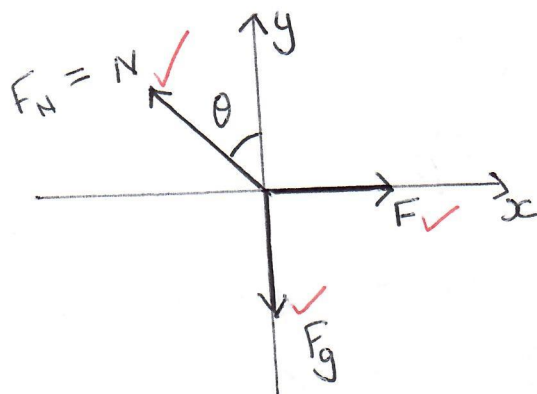
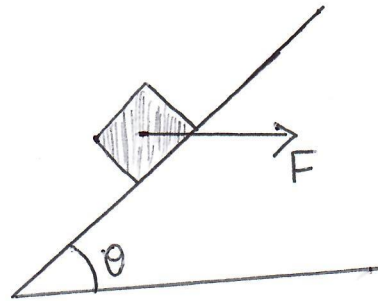
(a) Wat is die grootte van $\vec{F}$.

(b) Wat is die grootte van die krag wat die krat ondervind van die skuinsvlak?

(10)



Tutorial Test 4-1



NB (-1) if there is no reference frame / Axes.

$$\Sigma F_x = ma_x = 0 = F - N \sin \theta \quad \checkmark$$
$$F = N \sin \theta \quad \text{--- (1)}$$

$$\Sigma F_y = ma_y = 0 = N \cos \theta - F_g \quad \checkmark$$
$$N = F_g / \cos \theta \quad \text{--- (2)}$$

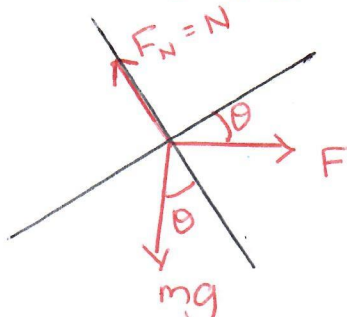
a) (2) in (1)

$$F = (F_g / \cos \theta) (\sin \theta) = F_g \tan \theta \quad \checkmark$$
$$= mg \tan \theta = (100 \text{ kg})(9.8 \text{ m/s}^2) \tan 30^\circ \quad \checkmark$$
$$= 566 \text{ N} \quad \checkmark$$

b)

$$N = F_g / \cos \theta \quad \checkmark = \frac{100 \text{ kg}(9.8 \text{ m/s}^2)}{\cos 30} \quad \checkmark$$
$$= 1.1 \times 10^3 \text{ N} \quad \checkmark$$

Can also use tilted axes.

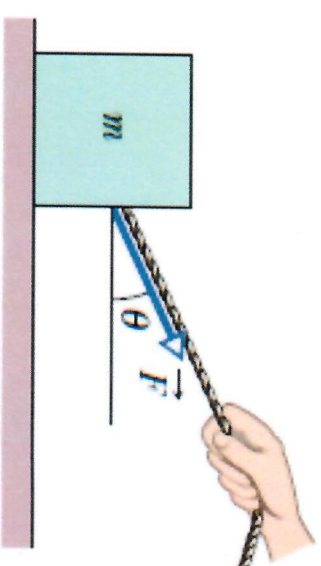


1. In Fig. 5-45, a block of mass $m = 5.00$ kg is pulled along a horizontal frictionless floor by a cord that exerts a force of magnitude $F = 12.0$ N at an angle $\theta = 25.0^\circ$.

(a) What is the magnitude of the block's acceleration?

(b) The force magnitude F is slowly increased. What is its value just before the block is lifted (completely) off the floor? (Hint: What happens to the normal force at this time?)

(c) What is the magnitude of the block's acceleration just before it is lifted (completely) off the floor?



In die figuur, 'n blok met massa $m = 5.00$ kg word getrek oor 'n horisontale wrywinglose vloer dmv 'n koord wat 'n krag van grootte $F = 12.0$ N teen 'n hoek $\theta = 25.0^\circ$.

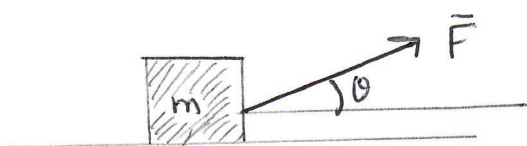
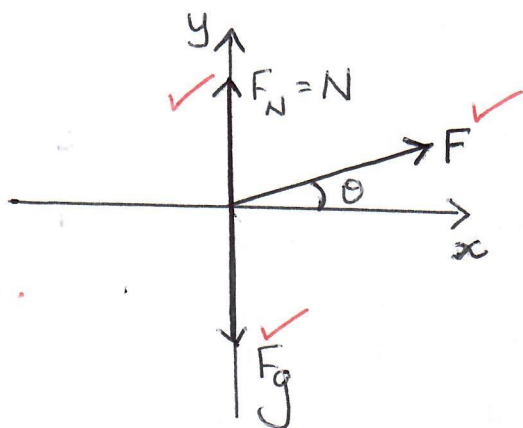
(a) Wat is die grootte van die blok se versnelling?

(b) Die grootte van die krag word gelydelik groter gemaak. Wat is die grootte van die krag net voordat die blok heeltemal opgelig word van die vloer af? (Wenk: Wat word van die normaalkrag op hierdie oomblik?)

(c) Wat is die grootte van die blok se versnelling net voordat dit van die vloer opgelig word?
(10)

Tutorial Test 4-2

① if there are no axes.



$$m = 5.00 \text{ kg}$$

$$F = 12.0 \text{ N}$$

$$\theta = 25^\circ$$

$$a) \Sigma F_x = ma_x = F \cos \theta \quad \text{--- ①}$$

$$a = \frac{F \cos \theta}{m} = \frac{12.0 \text{ N} \cos 25^\circ}{5.00 \text{ kg}}$$

$$= 2.18 \text{ m/s}^2$$

$$b) \Sigma F_y = ma_y = 0$$

$$= N + F \sin \theta - mg$$

When the block just lifts off the floor }
 $N = 0 \text{ N}$

$$\therefore \Sigma F_y = ma_y = 0 = F \sin \theta - mg$$

$$F = \frac{mg}{\sin \theta}$$

$$= \frac{5.00 \text{ kg} (9.8 \text{ m/s}^2)}{\sin 25^\circ} = 116 \text{ N}$$

c) Take $F = 116 \text{ N}$ (from (b)) and find a from ①

$$\Sigma F_x = ma_x = F \cos \theta$$

$$a_x = \frac{F \cos \theta}{m} = \frac{116 \text{ N} \cos 25^\circ}{5.00 \text{ kg}}$$

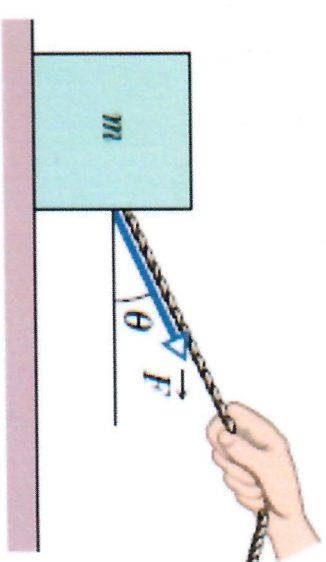
$$= 21.0 \text{ m/s}^2$$

1. In the figure, a block of mass $m = 5.00$ kg is pulled along a horizontal frictionless floor by a cord that exerts a force of magnitude F at an angle $\theta = 25.0^\circ$.

(a) What is the maximum force that can be exerted before the block lifts off the floor? (Hint: What happens to the normal force at this time?)

(b) What is the magnitude of the acceleration at this (a) time?

(c) Determine the normal force experienced by the block when the applied force is 15 N.



In die figuur, 'n blok met massa $m = 5.00$ kg word getrek oor 'n horisontale wrywinglose vloer dmv 'n koord wat 'n krag van grootte F teen 'n hoek $\theta = 25.0^\circ$ uitoefen op die blok.

(a) Wat is die maksimum krag wat uitgeoefen kan word voordat die blok van die vloer oplig?

(Wenk: Wat word van die normaal krag op hierdie oomblik?)

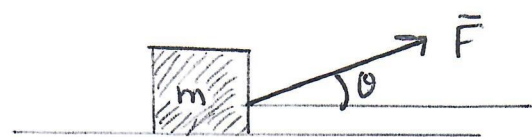
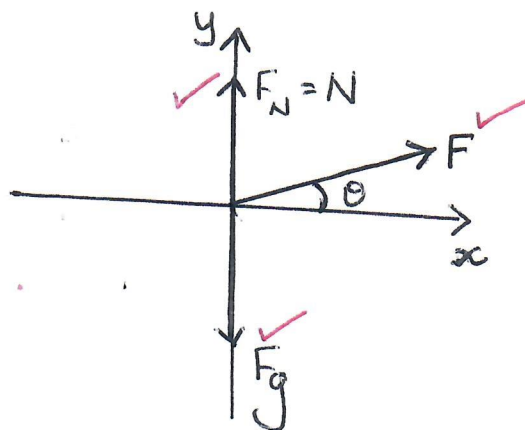
(b) Wat is die grootte van die versnelling van die blok op hierdie oomblik in (a).

(c) Bepaal die normaal krag op die blok wanneer die toegepaste krag 15 N is.

(10)

Tutorial Test 4-3

① if there are no axes.



$$m = 5.00 \text{ kg}$$

F.

$$\theta = 25^\circ$$

$$\Sigma F_y = ma_y = 0 = F_N + F \sin \theta - mg$$

a) As it lifts off the floor, $F_N = 0$, then

$$\Sigma F_y = ma_y = 0 = F_{\max} \sin \theta - mg$$

$$F_{\max} = mg / \sin \theta = \frac{5.00 \text{ kg} (9.8 \text{ m/s}^2)}{\sin 25^\circ}$$

$$= 116 \text{ N}$$

b) At this time $F = 116 \text{ N}$

$$\Sigma F_x = ma_x = F \cos \theta$$

$$a_x = \frac{F \cos \theta}{m} = \frac{116 \text{ N} \cos 25^\circ}{5.00 \text{ kg}}$$

$$= 21.0 \text{ m/s}^2$$

$$c) \Sigma F_y = ma_y = 0 = F_N + F \sin \theta - mg$$

$$F_N = mg - F \sin \theta$$

$$= 5.00 \text{ kg} (9.8 \text{ m/s}^2) - 116 \text{ N} \sin 25^\circ$$

$$= 42.7 \text{ N}$$